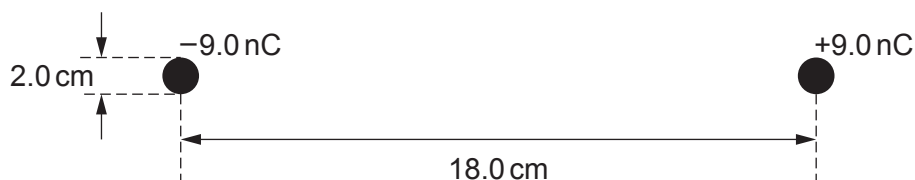


6. Two small spheres of radius 1.0 cm carry uniform charges of $+9.0 \text{ nC}$ and -9.0 nC . Their centres are separated by a distance of 18.0 cm as shown. Treat the charges as being at the centres of the spheres.



- (a) (i) Show that the potential at the surface of the negative sphere is approximately -7500 V . [3]

.....

.....

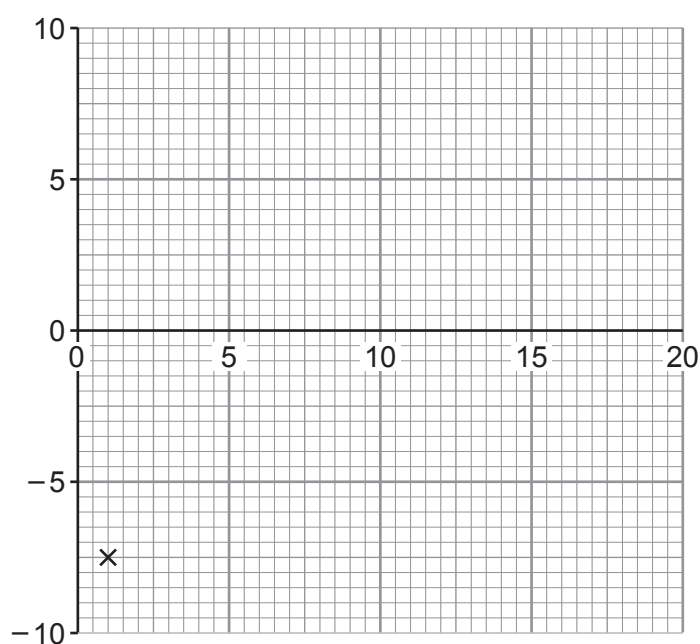
.....

.....

.....

- (ii) Sketch the graph for the electric potential between the two charges (the potential has been plotted for the surface of the negative charge). [3]

Potential / kV



Distance from
centre of
negative
charge / cm



- (b) Explain why the electric field on the right-hand surface of the negative sphere is approximately $810\,000\text{ NC}^{-1}$ and directed towards the left.

[3]

.....

.....

.....

.....

.....

9

1420U401
17